

Dust clustering and drifting in Keplerian turbulence

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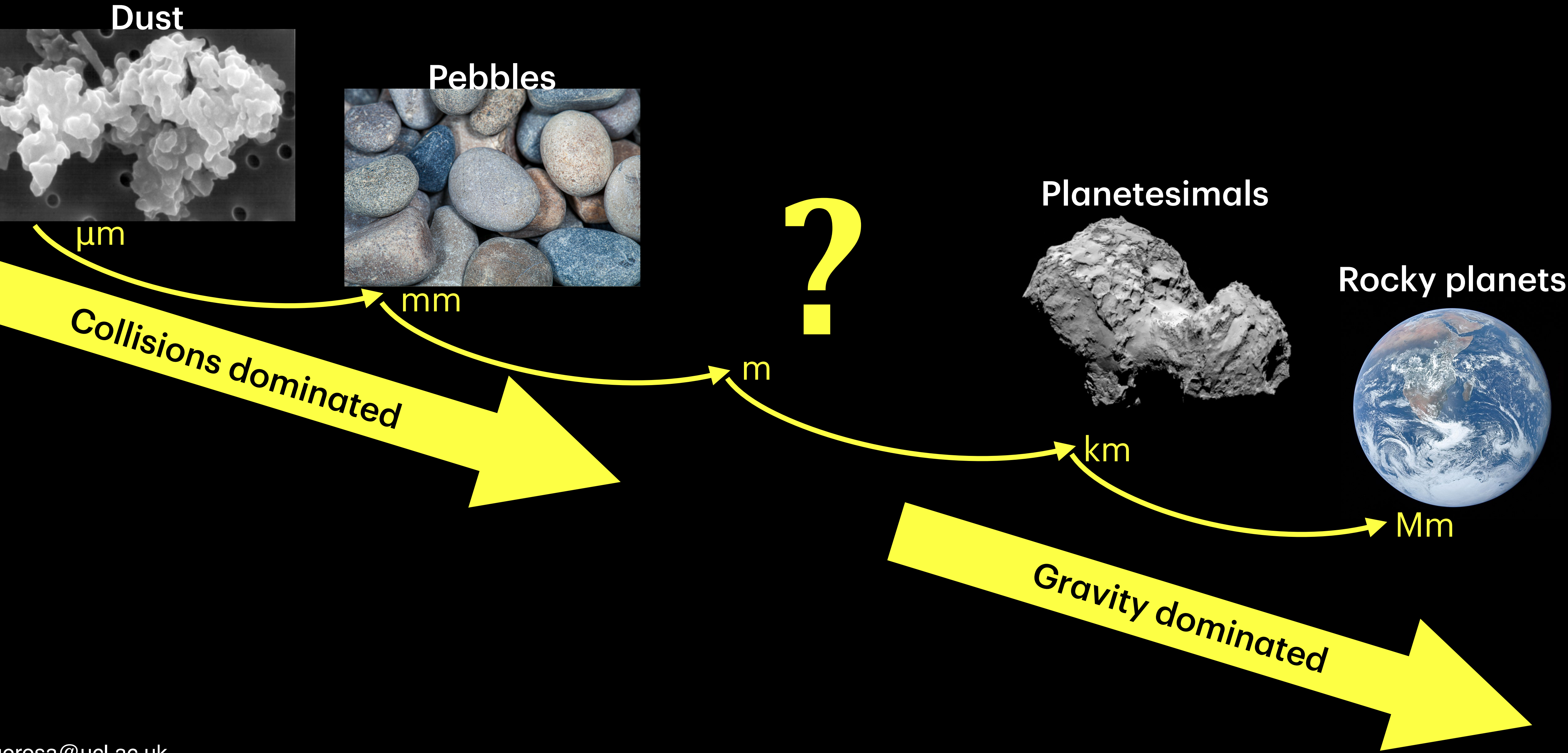
With Héloïse Méheut and Jérémie Bec

10 June 2026

3rd PFITS+ Meeting



Introduction



Introduction

Barriers to planetesimal formation:

- Fragmentation and radial drift
- Particularly important for m-sized aggregates

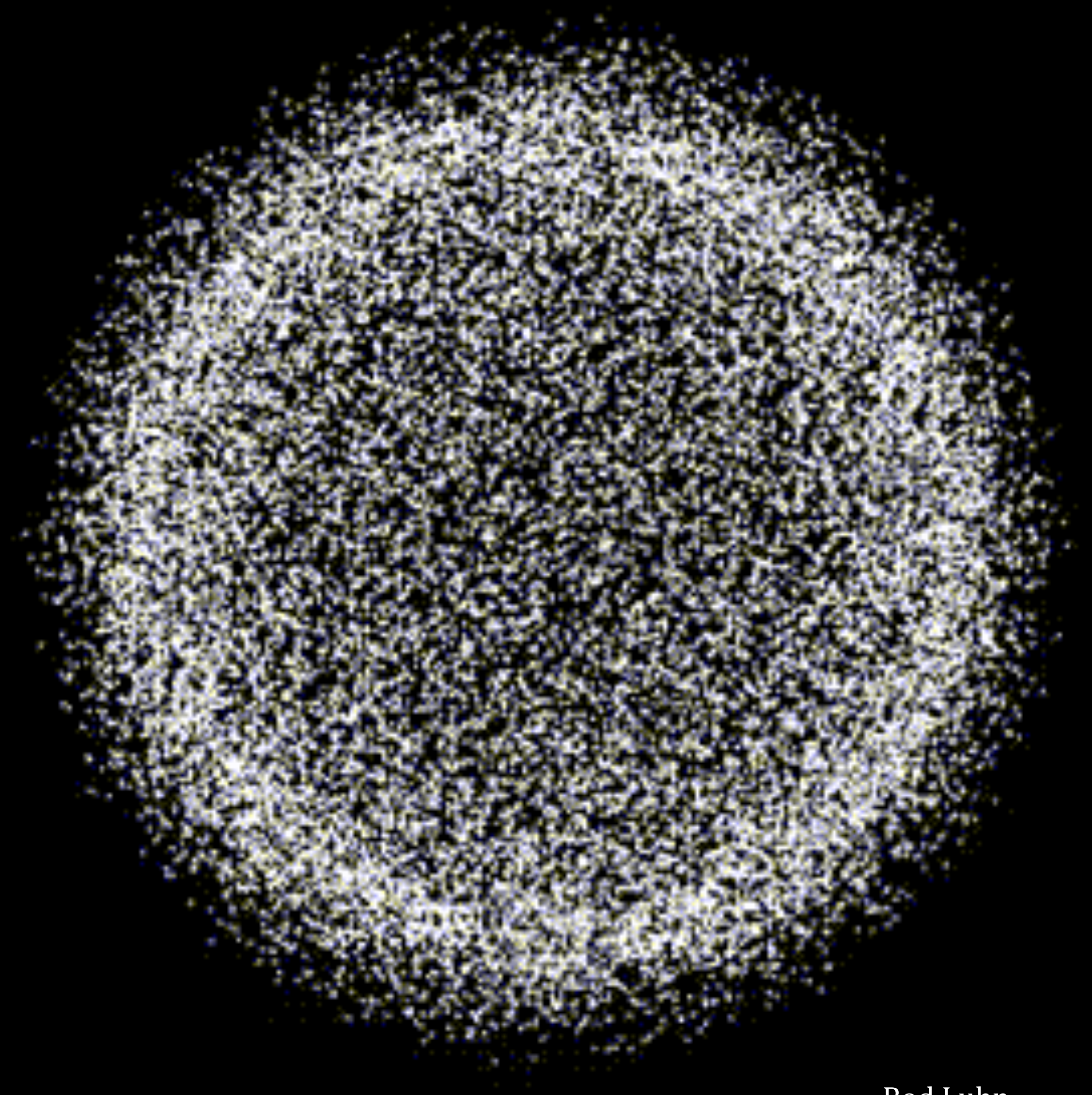
Overcoming the barriers:

Gravitational collapse

Efficient and fast way to form planetesimals

How to concentrate dust?

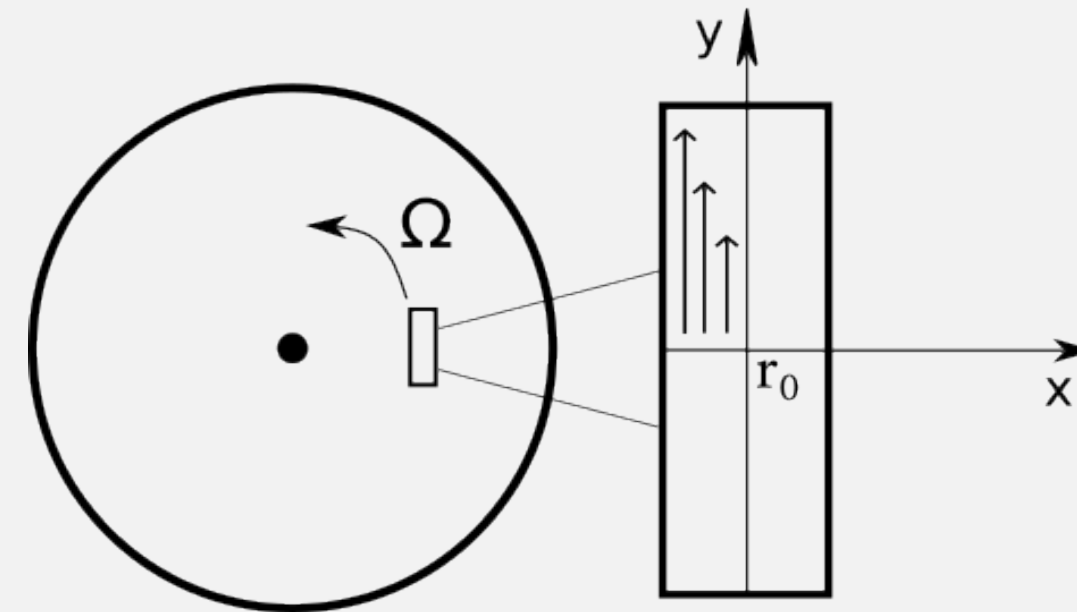
- Streaming instability (Youdin and Goodman, 2005)
- Dust concentration by turbulence



Direct numerical simulations

- ShearPart code (in-house)
- Snoopy code (Lesur and Longaretti, 2005)

- Shearing box approach



- Incompressible Navier-Stokes equation

$$\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{v} - 2\boldsymbol{\Omega} \times \mathbf{v} + 3\Omega^2 x \mathbf{e}_x + \mathbf{f}$$

$\nabla \cdot \mathbf{v} = 0$

Classical Navier-Stokes Coriolis force Centrifugal + gravitational force Forcing

- Shear rate parameter: $\sigma = \frac{3}{2} \Omega \tau_f$

Direct numerical simulations

→ Lagrangian approach

$$\frac{d\mathbf{R}_p}{dt} = \mathbf{V}_p$$

$$\frac{d\mathbf{V}_p}{dt} = -\frac{1}{\tau_p} [\mathbf{V}_p - \mathbf{v}(\mathbf{R}_p, t)] - 2\boldsymbol{\Omega} \times \mathbf{V}_p +$$

$$+ (2\Omega_K^2 + \Omega^2) R_{p,x} \mathbf{e}_x - (\Omega_K^2 - \Omega^2) r_0 \mathbf{e}_x$$

$\tau_p = \frac{\rho_p a_p}{\rho v_{th}}$

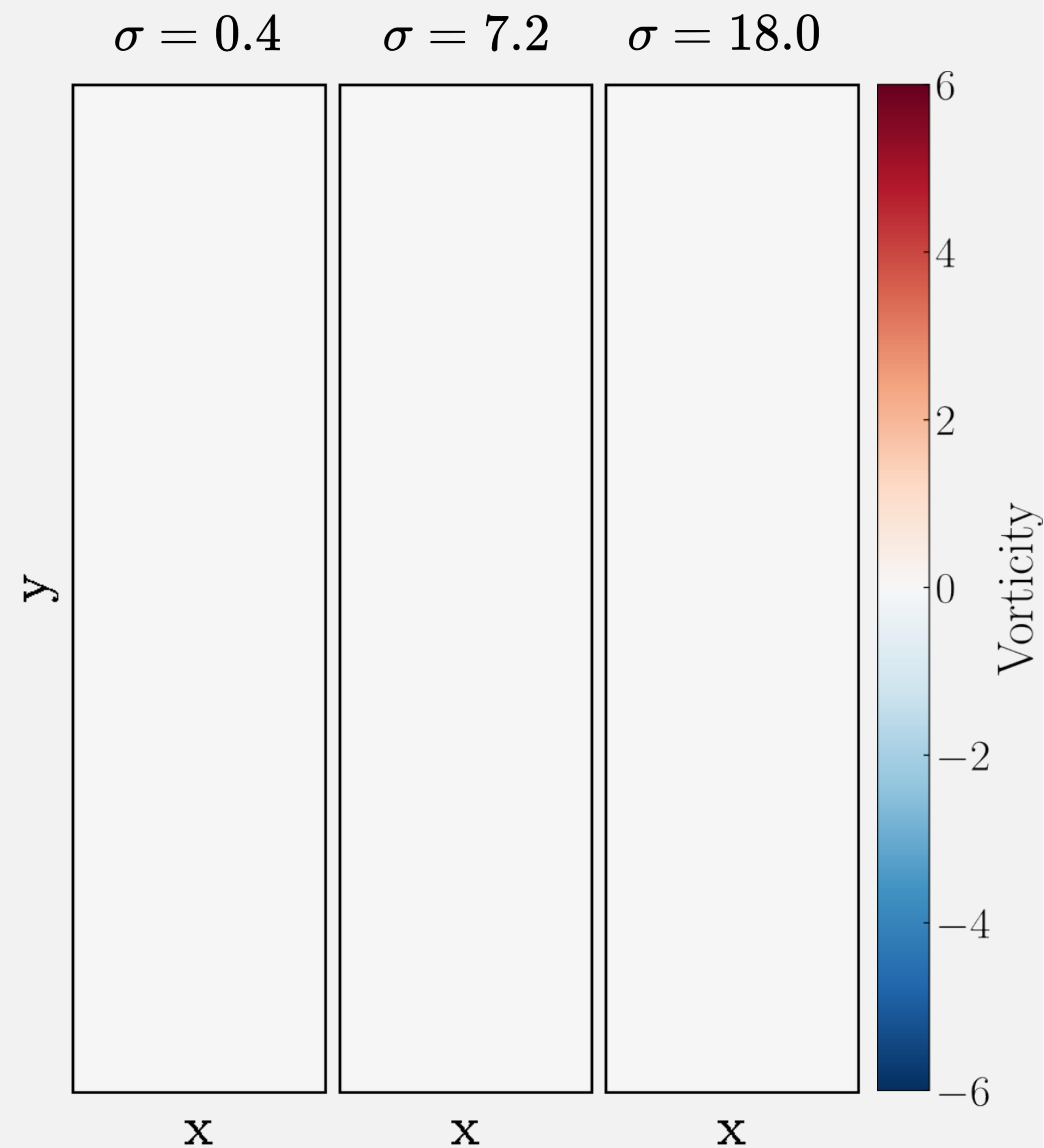
Gas drag
 Coriolis force
 Dust
 Gas
 Centrifugal + gravitational force
 Drift term

The diagram shows the equations of motion for a particle in a rotating frame. The first equation is $\frac{d\mathbf{R}_p}{dt} = \mathbf{V}_p$. The second equation is $\frac{d\mathbf{V}_p}{dt} = -\frac{1}{\tau_p} [\mathbf{V}_p - \mathbf{v}(\mathbf{R}_p, t)] - 2\boldsymbol{\Omega} \times \mathbf{V}_p + (2\Omega_K^2 + \Omega^2) R_{p,x} \mathbf{e}_x - (\Omega_K^2 - \Omega^2) r_0 \mathbf{e}_x$. The terms are color-coded and labeled: the first term in the second equation is in a yellow oval and labeled 'Gas drag'; the second term is in a purple oval and labeled 'Coriolis force'; the third term is in a blue oval and labeled 'Dust'; the fourth term is in a green oval and labeled 'Drift term'. The definition of τ_p is given as $\tau_p = \frac{\rho_p a_p}{\rho v_{th}}$. The labels 'Gas' and 'Centrifugal + gravitational force' point to the terms $\mathbf{v}(\mathbf{R}_p, t)$ and $(2\Omega_K^2 + \Omega^2) R_{p,x} \mathbf{e}_x$ respectively.

→ Drift parameter: $\epsilon = (\Omega_K^2 - \Omega^2) r_0$

→ Stokes number: $St_\omega = \tau_p / \tau_\omega$

Dust clustering in turbulence

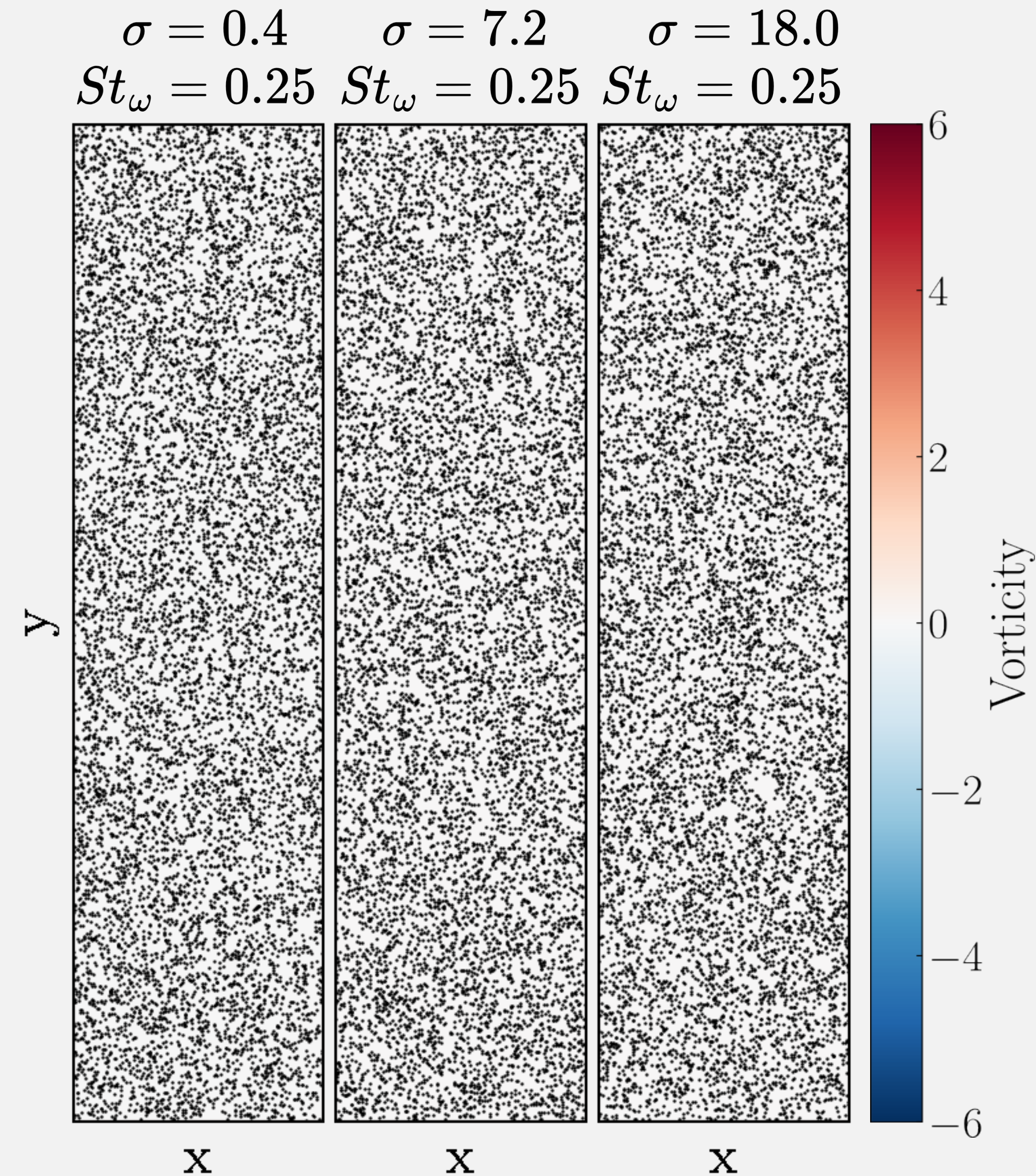


Gerosa et al., 2023

Dust clustering in turbulence

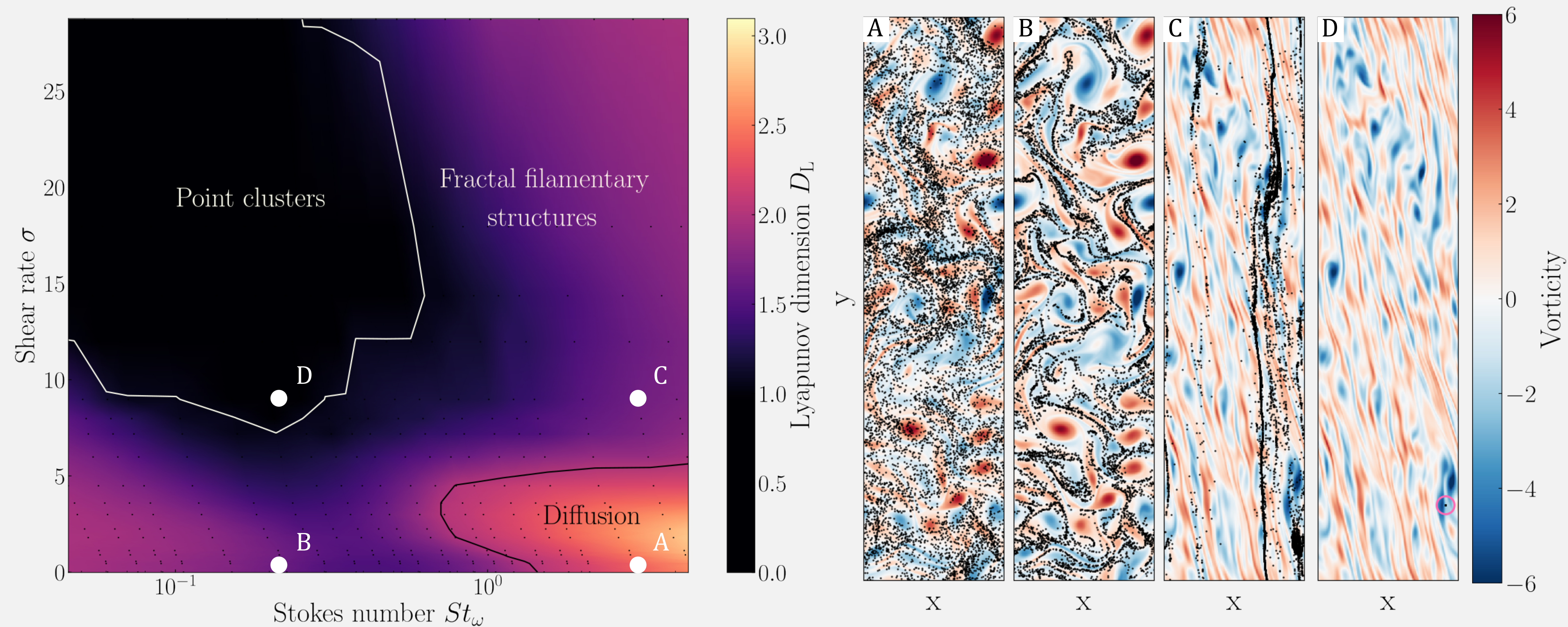
At large shear rates:

- Dust spends more time inside anticyclones
- Dust gets more clustered



Gerosa et al., 2023

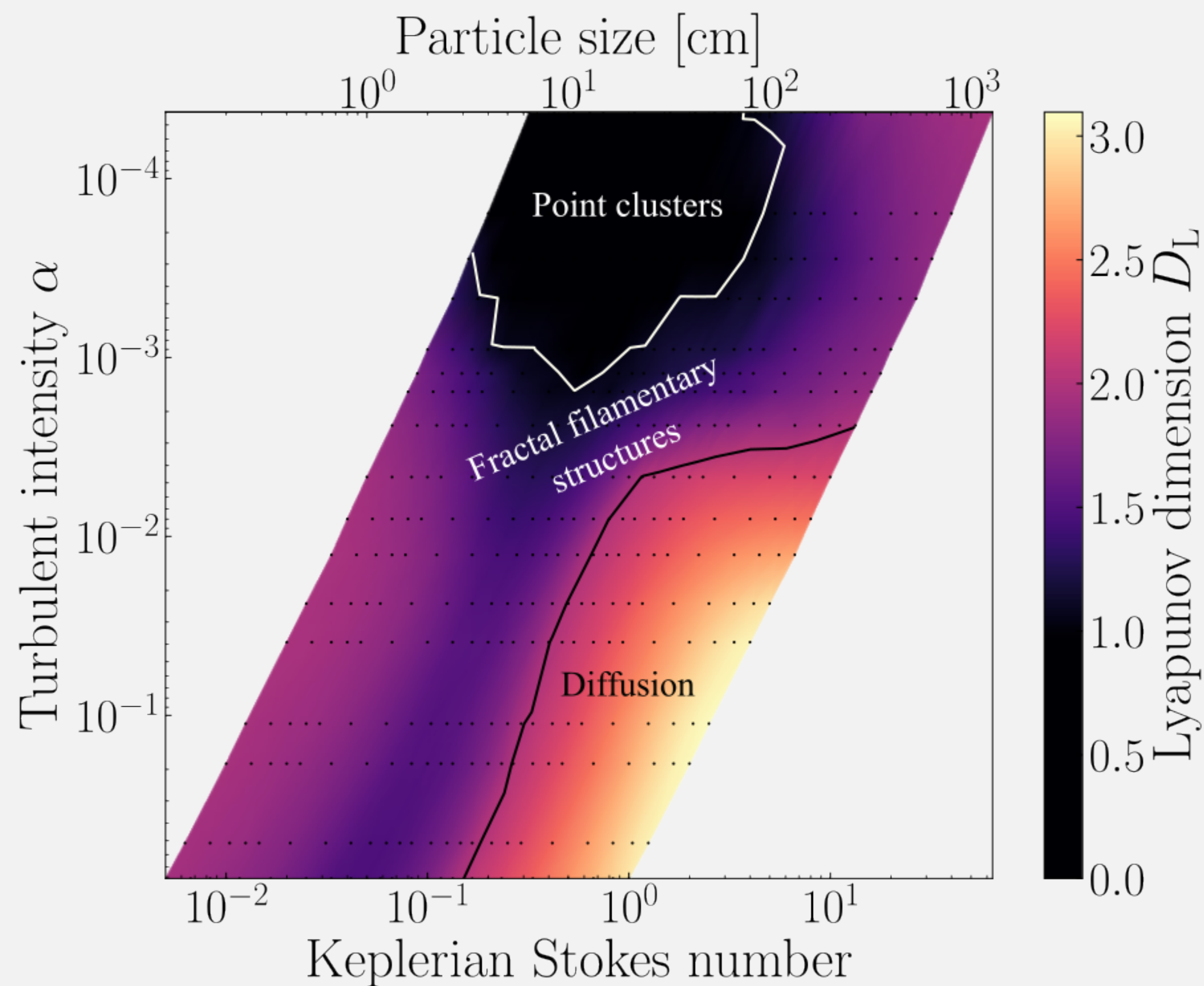
Dust clustering in turbulence



Gerosa et al., 2023

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Dust clustering in turbulence

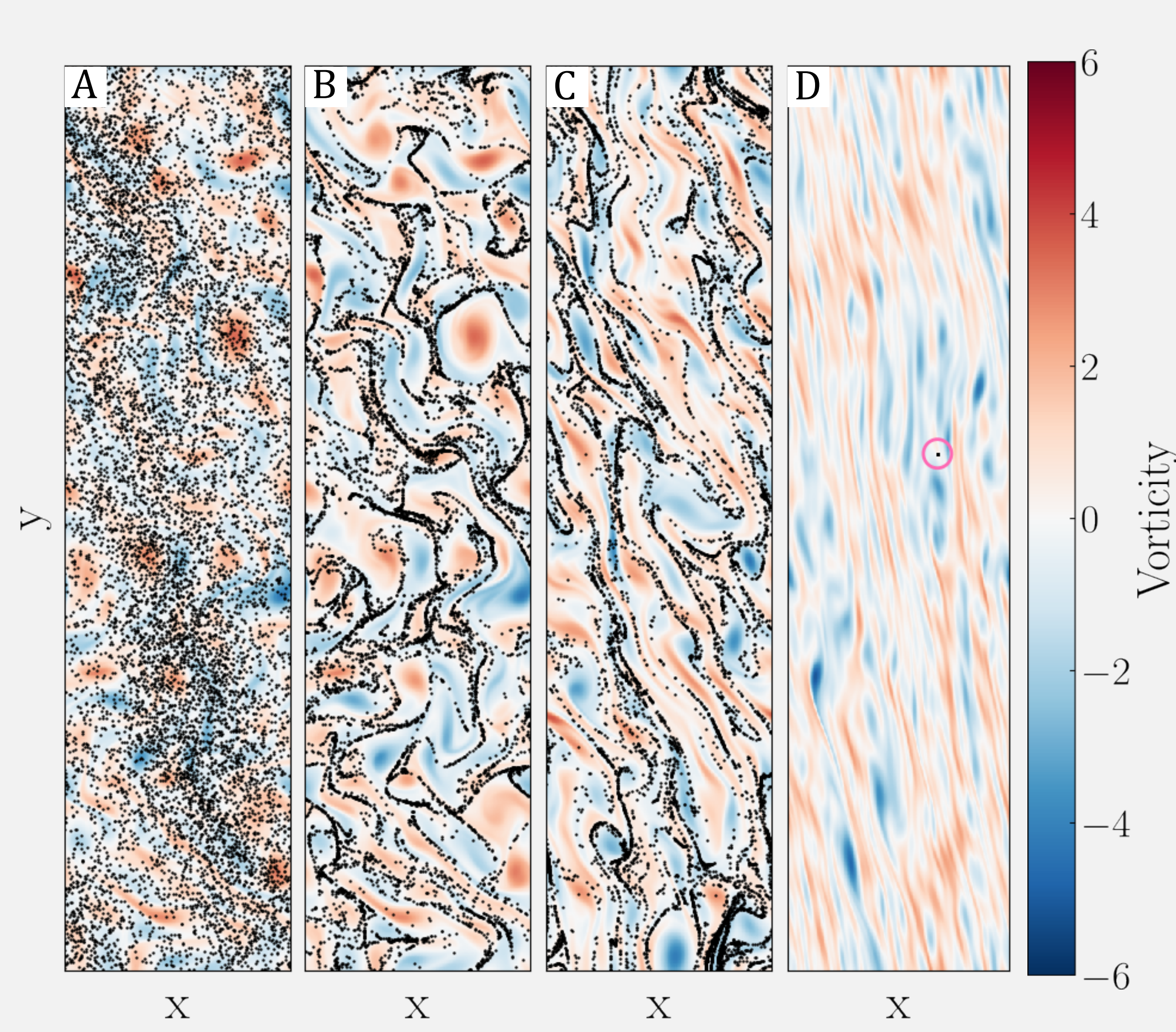
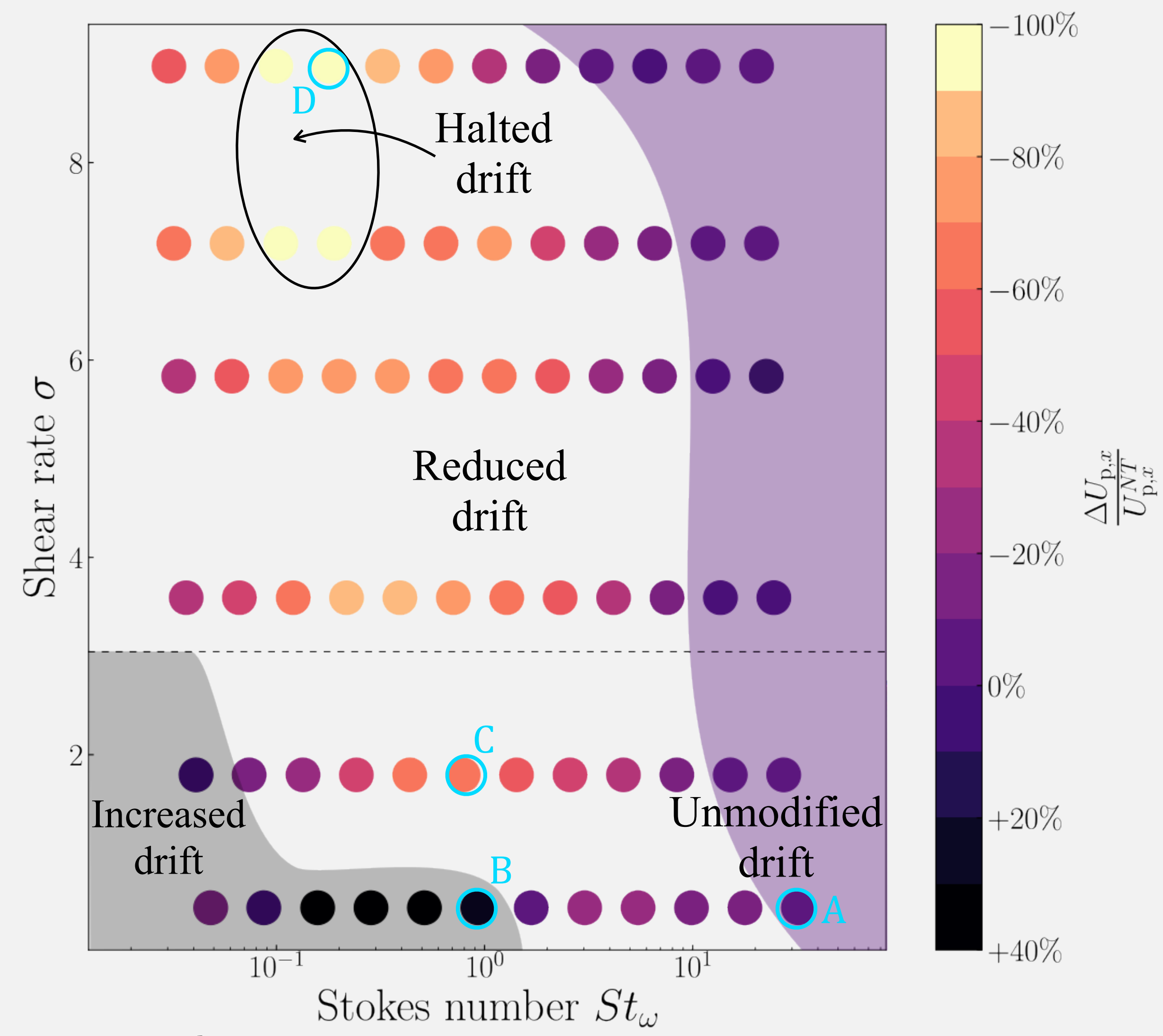


→ Shakura-Sunyaev
parameter:
(Shakura and Sunyaev, 1973)

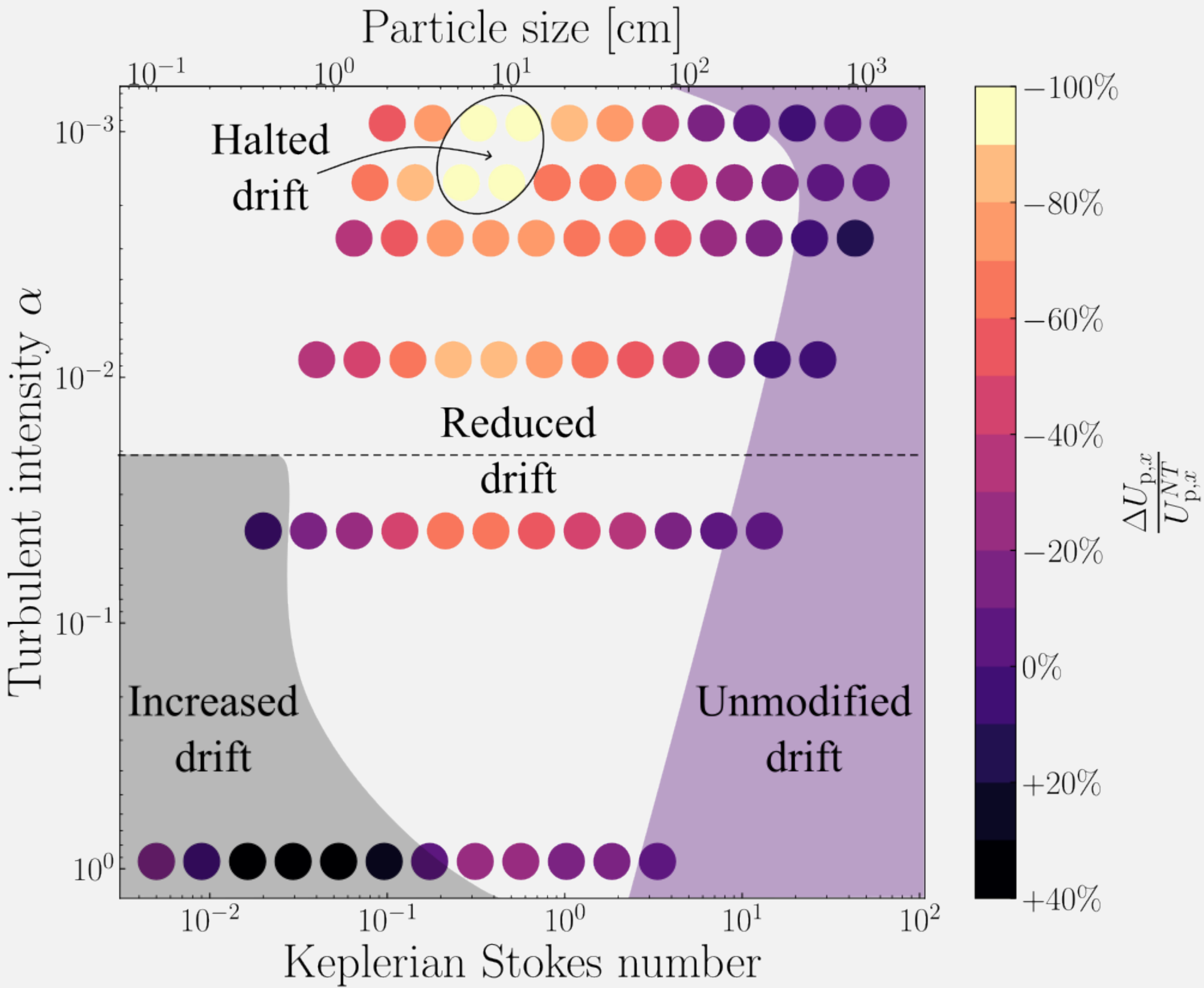
$$\alpha = \frac{\langle |\mathbf{v}'|^2 \rangle}{(\Omega H)^2}$$

Weak turbulence
strongly clusters dust

Dust drift in turbulence



Dust drift in turbulence

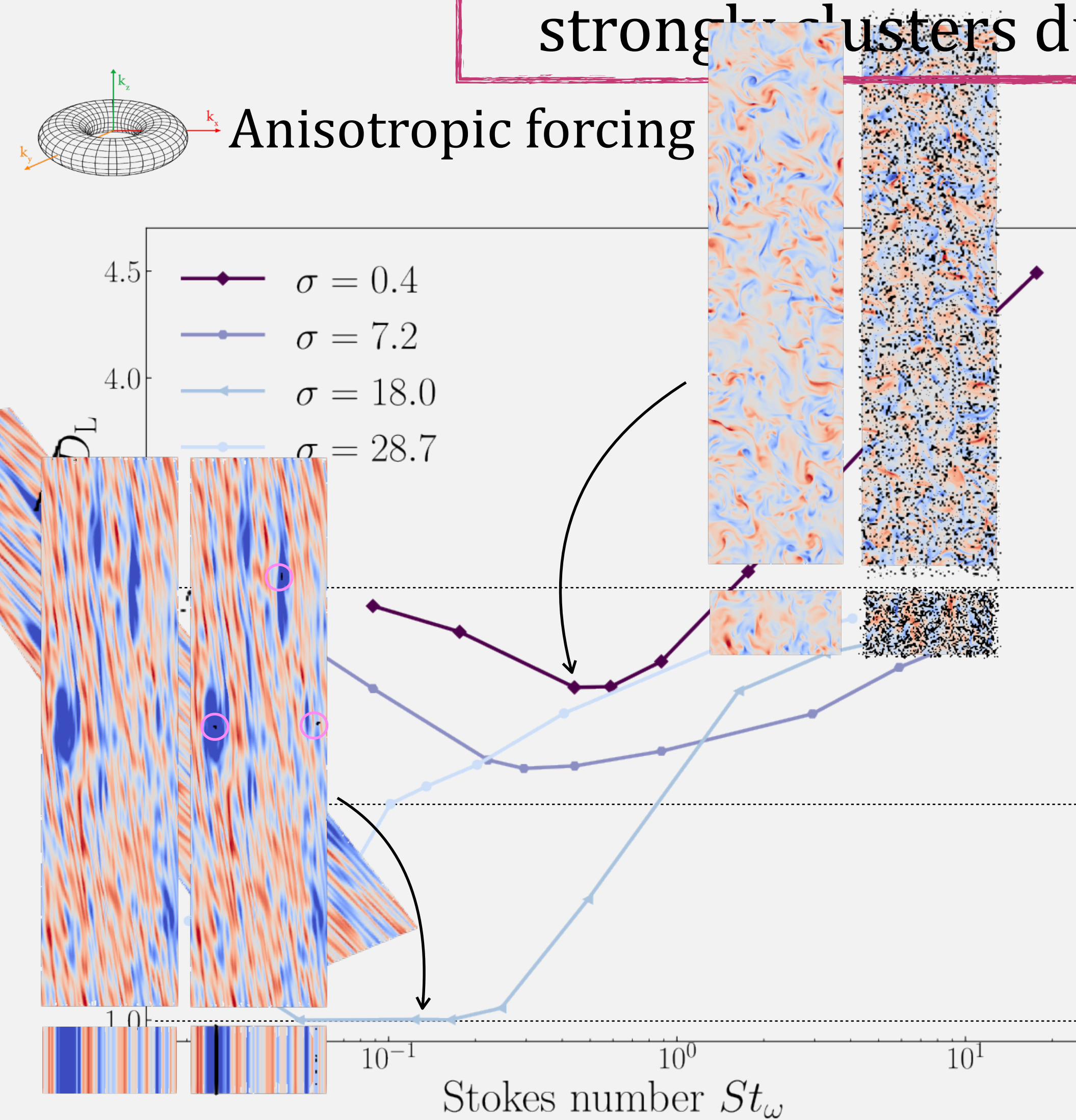
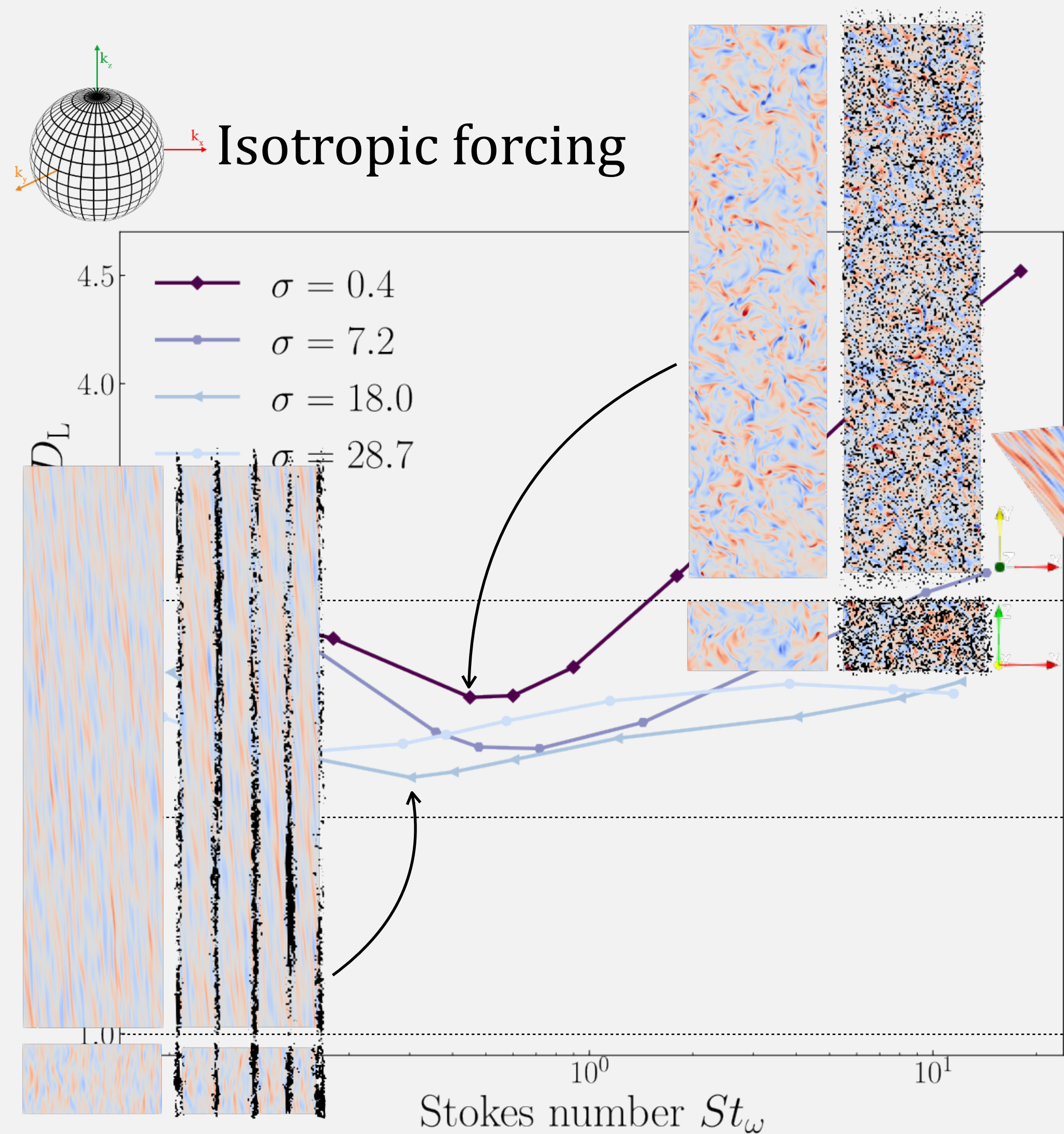


→ Shakura-Sunyaev parameter: $\alpha = \frac{\langle |\mathbf{v}'|^2 \rangle}{(\Omega H)^2}$
(Shakura and Sunyaev, 1973)

Weak turbulence strongly reduces the radial drift of dust

Dust clustering in 3D turbulence

3D results confirm that
weak turbulence
strongly clusters dust



Conclusions

